

RGB Hydra

CPU TIME LIMIT

1 second

MEMORY LIMIT

1024 MB

DIFFICULTY

medium (3.19999999814)

While hunting monsters, the brave hunter encounters a Hydra. But this was not just any Hydra - this is an RGB Hydra! This terrifying creature can have heads of three different colors. Initially, the Hydra has:

- R red heads,
- G green heads, and
- B blue heads.



Each time the hunter cuts off a head of a particular color, two new heads instantly grow back - one of each of the other two colors. For example, if the hunter cuts off a red head, then a green and a blue head will grow back immediately.

Driven by courage and ambition, the hunter not only wants to defeat the Hydra but also leave a lasting trace of the journey. To do this, the hunter wants as much red as possible to be visible on the Hydra - the goal is to maximize the number of red heads!

Before the Hydra launches its attack, the hunter has enough time to make at most k moves, where each move involves cutting off a single head.

What is the maximum number of red heads the Hydra can have after at most k moves?

Input

The input consists of a single line containing four integers R , G , B , and k ($0 \leq R, G, B, k \leq 10^{18}$), where:



- R is the number of red heads,
- G is the number of green heads,
- B is the number of blue heads,
- k is the maximum number of moves the hunter can perform.

It is guaranteed that the Hydra has at least one head initially, i.e., $R + G + B \geq 1$.

(Note that the numbers R, G, B, k are very large, and in languages like C++, you will need to use `long long` instead of `int`.)

Output

Print a single integer: the maximum number of red heads the Hydra can have after at most k moves.

Scoring

Your solution will be tested on a set of test groups, each worth a number of points. Each test group contains a set of test cases. To get the points for a test group you need to solve all test cases in the test group.

Group	Points	Constraints
1	20	$R, G, B, k \leq 10$
2	15	$k \leq R, G, B$
3	10	$R = 0$
4	10	$G = 0$
5	10	$B = 0$
6	35	No additional constraints.

Explanation of Samples

The Hydra initially has only 1 blue head, and the hunter can make at most 1 move. The only move that can be made is to cut off the blue head, which results in 1 red head and 1 green head. No more moves can be performed, which means that 1 red head is the maximum number.

The Hydra has 4 red heads, 3 green heads, and 2 blue heads, and the hunter can make at most 1 move. The hunter can, for example, choose to cut off a green head, which results in 5 red heads, 2 green heads, and 3 blue heads. After this move, no further moves can be performed. There is no way to get more than 5 red heads, so the answer is 5.

The Hydra has 10 heads of each color. Since the hunter cannot make any moves (as $k = 0$), the number of red heads remains 10.

Sample Input 1

```
0 0 1 1
```

Sample Output 1

```
1
```

Sample Input 2

```
4 3 2 1
```

Sample Output 2

```
5
```

Sample Input 3

```
10 10 10 0
```

Sample Output 3

```
10
```

Sample Input 4

```
926332030620250124  
57836736120250125  
503852839920250126  
804672315982687419
```

Sample Output 4

```
1731004346602937543
```